1(a)	$h = 0.5$		B1
	$A \approx \frac{1}{2} \times \frac{1}{2}\{6.792 + 5.113 + 2(6.298 + 5.858 + 5.466)\}$ Correct application of the trapezium rule with their h		M1
	= 11.79		A1
			(3)
(b)(i)	$A \approx 2 \times "11.79"$		M1
	= 23.58		A1ft
(b)(ii)	$A \approx "11.79" + 6$		M1
	= 17.79		A1ft
			(4)
			Total 7

2(a)	$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20$ $\Rightarrow \left(\frac{dy}{dx} =\right) \frac{27}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}}$	M1 A1
		(2)
(b)	$\frac{27}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow x = ...$	dM1
	$x = 9$	A1
	$x = 9 \Rightarrow y = ...$	M1
	$y = 34$	A1
		(4)
(c)	$\frac{d^2y}{dx^2} = \pm Ax^{-\frac{3}{2}} \pm Bx^{-\frac{1}{2}}$ $\left(\frac{d^2y}{dx^2}\right)_{x=9} = -\frac{27}{4}(9)^{-\frac{3}{2}} - \frac{3}{4}(9)^{-\frac{1}{2}} \left(= -\frac{1}{2}\right)$	M1
	$\frac{d^2y}{dx^2} = -\frac{27}{4}x^{-\frac{3}{2}} - \frac{3}{4}x^{-\frac{1}{2}}$ $\frac{d^2y}{dx^2} < 0$ for x > 0 / when x = 9 so maximum	A1
		(2)
		Total 8

3(a)	$\left(2-\frac{kx}{4}\right)^8 = 2^8 + \binom{8}{1}2^7\left(-\frac{kx}{4}\right) + \binom{8}{2}2^6\left(-\frac{kx}{4}\right)^2 + \binom{8}{3}2^5\left(-\frac{kx}{4}\right)^3 + \ldots$ Or $\left(1-\frac{kx}{8}\right)^8 = 1 + \binom{8}{1}\left(-\frac{kx}{8}\right) + \binom{8}{2}\left(-\frac{kx}{8}\right)^2 + \binom{8}{3}\left(-\frac{kx}{8}\right)^3 + \ldots$	M1
	$= 256 - 256kx + 112k^2x^2 - 28k^3x^3 + \ldots$	B1 A1 A1 **(4)**
(b)	$f(x) = (5-3x)\left(2-\frac{kx}{4}\right)^8 = (5-3x)\left(256 - 256kx + 112k^2x^2 - 28k^3x^3 + \ldots\right)$ Coefficient of x is $5\times -256k - 3\times 256$	M1
	$5\times 256 = 3(-1280k - 768) \Rightarrow k = \ldots$	M1
	$k = -\frac{14}{15}$	A1 **(3)**
		Total 7

4	$2\log_3(1-x) = \log_3(1-x)^2$ or $3 = \log_3 3^3$	B1
	$\log_3(32-12x) - \log_3(1-x)^2 = \log_3 \dfrac{32-12x}{(1-x)^2}$ Combines 2 log terms correctly	M1
	$\dfrac{32-12x}{(1-x)^2} = 27$	A1
	$\Rightarrow 27x^2 - 42x - 5 = 0 \Rightarrow x = \ldots$	M1
	$x = -\dfrac{1}{9}$	A1
		(5)
		Total 5

5(a)	$3(1)^3 + A(1)^2 + B(1) - 10 = k$ **or** $3(-1)^3 + A(-1)^2 + B(-1) - 10 = -10k$	M1
	$A + B - 7 = k,\ A - B - 13 = -10k$ $\Rightarrow -10A - 10B + 70 = A - B - 13$	M1
	$\Rightarrow 11A + 9B = 83$*	A1*
		(3)
(b)	$3\left(\frac{2}{3}\right)^3 + A\left(\frac{2}{3}\right)^2 + B\left(\frac{2}{3}\right) - 10 = 0$	M1
	$11A + 9B = 83,\ 12A + 18B = 246$ $\Rightarrow A = ..., B = ...$	M1
	$A = -8, B = 19$	A1
		(3)
(c)	$\mathrm{f}(x) = (3x - 2)(x^2 + ...x + ...)$	M1
	$g(x) = x^2 - 2x + 5$	A1
		(2)
		Total 8

6(a)	Examples: $m_{PQ} = \frac{14+30}{23-15},\ m_{QR} = \frac{-30+26}{15+7} \Rightarrow m_{PQ} \times m_{QR} = ..$ or $PQ^2 = 8^2 + 44^2, QR^2 = 22^2 + 4^2,\ PR^2 = 30^2 + 40^2$ $PQ^2 + QR^2 = ..$	M1
	$m_{PQ} \times m_{QR} = \frac{11}{2} \times \left(-\frac{2}{11}\right) = -1 \Rightarrow \angle PQR = 90°$ Or $PQ^2 = 2000,\ QR^2 = 500,\ PR^2 = 2500$ $2000 + 500 = 2500 \Rightarrow \angle PQR = 90°$	A1
		(2)
(b)(i)	Centre is $(8, -6)$	B1
(b)(ii)	$r = \sqrt{(23-8)^2 + (14+6)^2}$ or e.g. $r = \frac{1}{2}\sqrt{(23+7)^2 + (14+26)^2}$	M1
	$r = 25$	A1
		(3)
(c)	S is $(1, 18)$ **or** $m_T = \frac{7}{24}$	B1
	$m_N = \frac{18+6}{1-8} \Rightarrow m_T = \frac{8-1}{18+6} \Rightarrow y - 18 = \frac{7}{24}(x-1)$ Uses a correct straight line method for the tangent using their S and the negative reciprocal of the radius gradient	M1
	$7x - 24y + 425 = 0$	A1
		(3)
		Total 8

7(i)	$3\sin(2x-15°)=\cos(2x-15°)$ $\Rightarrow \tan(2x-15°)=\frac{1}{3}$	M1
	$\Rightarrow x=\frac{\tan^{-1}\left(\frac{1}{3}\right)\pm 15°}{2}$	M1
	$x=16.7°,\ -73.3°$	A1 A1
		(4)
(ii)	$4\sin^2\theta+8\cos\theta=3$ $\Rightarrow 4\left(1-\cos^2\theta\right)+8\cos\theta=3$ $\Rightarrow 4\cos^2\theta-8\cos\theta-1=0$	M1
	$\cos\theta=\frac{8\pm\sqrt{64+4\times 4}}{2\times 4}\Rightarrow\theta=\cos^{-1}\left(1-\frac{\sqrt{5}}{2}\right)=...$	M1
	$\theta=1.69,\ 4.59$	A1 A1
		(4)
		Total 8

8(a)	$u_{20} = 100 + 19(-2) = 62$*	B1*
		(1)
(b)	$S_{20} = \frac{1}{2}(20)\{2\times 100 + 19(-2)\} = \ldots$ **or** $S_{20} = \frac{1}{2}(20)\{100 + 62\} = \ldots$	M1
	$= 1620$ (mm)	A1
		(2)
(c)	$62 \times r^2 = 60 \Rightarrow r^2 = \ldots$	M1
	$r^2 = \frac{60}{62} \Rightarrow r = \sqrt{\frac{60}{62}}$ = 0.983738...	A1
		(2)
(d)	Total distance from GS hits = $\frac{62 \times 0.983\ldots\left(1 - 0.983\ldots^{n}\right)}{1 - 0.983\ldots}$	M1
	$1620 + \frac{62 \times 0.983\ldots\left(1 - 0.983\ldots^{n}\right)}{1 - 0.983\ldots} > 3000$	M1
	$0.983\ldots^{n} < 0.63207\ldots \Rightarrow n = \frac{\log(0.63207\ldots)}{\log(0.983\ldots)}$	M1
	$n = 27.98\ldots \Rightarrow N = 20 + 28 = 48$	A1cso
		(4)
	(d) Alternative taking 20th hit as first term of GP	
	Total distance from GS hits = $\frac{62\left(1 - 0.983\ldots^{n}\right)}{1 - 0.983\ldots}$	M1
	$1620 - 62 + \frac{62\left(1 - 0.983\ldots^{n}\right)}{1 - 0.983\ldots} > 3000$	M1
	$0.983\ldots^{n} < 0.62179\ldots \Rightarrow n = \frac{\log(0.62179\ldots)}{\log(0.983\ldots)}$	M1
	$n = 28.98\ldots \Rightarrow N = 19 + 29 = 48$	A1cso
		Total 9

9(a)	$mx = x - x^2 \Rightarrow m = 1 - x \Rightarrow x = ...$ Or $y = \frac{y}{m} - \frac{y^2}{m^2} \Rightarrow m^2 = m - y \Rightarrow y = ...$	M1
	$x = 1 - m$ **and** $y = m(1-m)$	A1
		(2)
(b)	$\int x - x^2(-mx)\,dx = \frac{x^2}{2} - \frac{x^3}{3}\left(-m\frac{x^2}{2}\right)$	M1
	$\text{Area of } R_1 = \int_0^{"1-m"} \{x - x^2(-mx)\}\,dx$ $= \frac{(1-m)^2}{2} - \frac{(1-m)^3}{3}\left(-m\frac{(1-m)^2}{2}\right) - 0$	M1 A1
	$\text{Area of } R_1 = \int_0^{"1-m"} \{x - x^2 - mx\}\,dx = \frac{(1-m)^2}{2}(1-m) - \frac{(1-m)^3}{3}(-0)$ Correct strategy for the area (may be scored for finding separate areas and subtracting)	dM1
	$= \frac{(1-m)^3}{6}$*	A1*
		(5)
(c)	$\text{Area of } (R_1 + R_2) = \int_0^1 (x - x^2)\,dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = ...$ $\left(= \frac{1}{6}\right)$	M1
	Alt: $\text{Area of } R_2 = \int_{1-m}^1 (x - x^2)\,dx + \frac{1}{2}("1-m") \times m(1-m)$ $= \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{1-m}^1 + \frac{1}{2}m(1-m)^2 = ...\left(= \frac{1}{6} - \right.$	
	$R_1 = R_2 \Rightarrow \frac{(1-m)^3}{6} = \frac{1}{12} \Rightarrow m = ...$	dM1
	Alt: $\frac{(1-m)^3}{6} = \frac{1}{6} - \frac{(1-m)^2}{2} + \frac{(1-m)^3}{3} + \frac{1}{2}m(1-m)^2 \Rightarrow m = ...$	
	$m = 1 - \frac{1}{\sqrt[3]{2}}$	A1
		(3)
		Total 10

10(i)	E.g. $p = 7 \Rightarrow 2p + 1 = 15$ Which is not a prime number (so the statement is not true)	B1
		(1)
(ii)	n odd $\Rightarrow n = 2k + 1$ $\Rightarrow 5n^2 + n + 12 = 5(2k+1)^2 + 2k + 1 + 12$ **or** n even $\Rightarrow n = 2k$ $\Rightarrow 5n^2 + n + 12 = 5(2k)^2 + 2k + 12$	M1
	n odd $\Rightarrow n = 2k + 1$ $\Rightarrow 5n^2 + n + 12 = 5(2k+1)^2 + 2k + 1 + 12$ **and** n even $\Rightarrow n = 2k$ $\Rightarrow 5n^2 + n + 12 = 5(2k)^2 + 2k + 12$	M1
	n odd: $20k^2 + 22k + 18$ which is even and n even: $20k^2 + 2k + 12$ which is even	A1
	n odd: $2(10k^2 + 11k + 9)$ and n even: $2(10k^2 + k + 6)$ These are both even so $5n^2 + n + 12$ must be even for all integers n	A1
		(4)
		Total 5